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(1) Research Program/Project/Study Title: Rehabilitation of Clonal Nursery and Asexual Propagation of Selected Fruits and Ornamental Plants	
Proponent Name:	Dr. Salvacion J. Legaspi
Academic Rank:	Prof. VI
Co-Authors (with academic rank):	Kient Jerwin V. Tungala
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E-mail Address:	

Program or Project has component (*if no proceed to 2*)? ☐ Yes ☐ No

Component Title:	
Component Leader:	
Academic Rank:	

(2) Category (please check) ☐ Social Science/Education /Humanities & other related fields ☒ Natural/BiologicalSciences ☐ Food Research/ ☐ Developmental/ Computer/Engineering/ Architecture ☐ Extension ☐ Poster	(3) Research Center [Check the concerned center] ☐ SSITRDC Social Science and Industrial Technology Research Development Center ☐ LRDC Livestock Research Development Center ☐ FFRDC Fisheries and Food Research Development Center ☒ CSRDC Crop Science Research Development Center
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(4) Site of implementation							
Municipality/City:	Mambusao						
Province:	Capiz						
Region:	Region VI						
(5) Classification <table> <tr> <td>Research:</td> <td>Development:</td> </tr> <tr> <td>☐ Basic</td> <td>☐ Pilot Testing</td> </tr> <tr> <td>☐ Applied</td> <td>☐ Tech. Promotion/Commercialization</td> </tr> </table>	Research:	Development:	☐ Basic	☐ Pilot Testing	☐ Applied	☐ Tech. Promotion/Commercialization	(6) Mode of Implementation ☐ Single Authorship/Proponent ☐ Collaborative Research (within campus) ☐ Collaborative Research (intercampus) ☐ Collaborative Research (external agency)
Research:	Development:						
☐ Basic	☐ Pilot Testing						
☐ Applied	☐ Tech. Promotion/Commercialization						
(7) Sector/Commodity	(8) Discipline/Field of Specialization						

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(9) Rationale (State the research problem and significance of the project to the current needs of the country. The proposal should justify resource expenditure. A typical justification would include a brief introduction, a general statement concerning the historical basis for R & D and, thrusts being addressed, utilization of the expected output and the impact information generated will have on science, the target users and the country.)

The Philippines boasts a rich diversity of native trees, with approximately 3,600 species identified, out of which 67% are exclusive to our archipelago (GCSFHESE 2012). Globally, there are around 2,500 tropical fruits, with over 300 edible perennial plant species documented in the Philippines. However, only five of these fruits, including banana, pineapple, mango, papaya, and citrus, are considered major, while another 20 species, such as avocado, breadfruit, caimito, and cashew, hold economic significance but are underutilized (Coronel 2011). These lesser-known fruit species show significant promise for local and international markets, boasting high nutritional value, versatile applications, potential for processing, and capacity for job creation within local communities, despite being classified as rare. (Coronel 1998).

Forests, as renewable natural resources, constitute ecosystems dominated by trees and woody plants (Park, 2006). They fulfill various human needs, including timber, food, fodder, fuel, medicine, and building materials, while also providing essential protection for both humans and animals, and enhancing soil fertility (Doran & Zeiss, 2000). However, deforestation poses a significant threat to the survival of natural forests and their benefits worldwide. To ensure their continued usefulness, forest sustainability relies on proper management, sustainable utilization, conservation practices, and effective artificial regeneration, often accomplished through seedling production (Park, 2006).

Yet, seed availability often presents challenges due to factors like infrequent fruiting, low seed production, dormancy, pests, diseases, and rapid seed viability loss (Park, 2006). Moreover, reliance on sexual reproduction through seeds poses risks of losing genetic characteristics (Henrique et al., 2006). Thus, exploring alternative regeneration methods, such as vegetative propagation, becomes imperative (Leakey, 2004). This method, which involves asexual reproduction to produce genetically identical offspring, offers a practical solution for large-scale propagation and preservation of desired genetic traits (Henrique et al., 2006). Vegetative propagation techniques include stem cuttings providing an alternative means for sustainable forest management and agricultural production, particularly for species with limited seed availability (Leakey, 2004).

Cutting propagation serves as an alternative technique employed instead of sexual propagation because it: firstly, streamlines the production process; secondly, decreases the time required for seedling production; thirdly, circumvents genetic variability; and fourthly, can be utilized for both rootstock and scion propagation (OLIVEIRA et al., 2003; CARDOSO et al., 2011).

Many important aspects related to the propagation of different varieties of fruit trees and ornamental plants through cuttings have not been studied in depth. One such aspect is the evaluation of the survival rates and growth performance of these plants. To address this gap in knowledge, our study aims to investigate how selected fruit trees and ornamental plants respond to clonal nursery conditions via cuttings. By exploring these uncharted territories, we intend to provide valuable insights into the subtle dynamics of cutting propagation. This will help us better understand its effectiveness in promoting sustainable agricultural practices.

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(9-A) Project Description (Discuss the project strategy and plan of action of the project. It should also describe the technical procedures and other works needed to carry out the project objectives. The relationship of the project inputs and activities should be explained clearly in this section. Projects similar/related to the proposed project should be mentioned. .)

(10) Objectives/Statement of the Problem (for Social Sciences) (Statement of the specific purpose to address the problem areas of the project.)

The study aims to determine the response of selected fruits and ornamental plants using asexual propagation in clonal nursery condition. Specifically, it will attempt:

1. To determine the survival rate of selected fruits and ornamental plants cuttings grown under clonal nursery;
2. To evaluate the response of selected fruit trees and ornamental plants on clonal propagation through cuttings;
3. To identify the optimal length and stages of stem development for initiating cuttings and determine its influence on the subsequent growth of selected fruit trees and ornamental plants:
4. To identify if there is an interaction effect between the length of cuttings and stages of stem of different fruits trees and ornamental plants.

(11) Review of Literature/Prior Art Search (Related researchers/activities that have been conducted. State-of –the-art of current technology / information from which the project proposal will take-off).

The cultivation of fruit typically commences with the propagation of fruit seedlings, which can be achieved through sexual or asexual methods. Sexual reproduction, involving seeds, is often avoided in fruit cultivation due to the risk of foreign pollination and the occurrence of dichogamy in certain fruit species. Asexual reproduction methods encompass various techniques such as apomictic embryo reproduction, layering, separation, cutting, and grafting (Kaşka ve Yılmaz, 1987). In horticulture, maintaining the desired traits of parent plants is crucial for successful plant reproduction. Depending on the species, different techniques are employed to optimize nursery production systems and address specific challenges. For instance, almond cultivation faces the challenge of non-infectious bud failure (NBF), necessitating the implementation of effective nursery practices and the careful selection of plant materials and methods (Roberto and Colombo, 2020).

Asexual reproduction involves the propagation of plants using specialized vegetative organs, such as roots, shoots, stems, or leaves, capable of forming new plants. This method ensures that the offspring are genetically identical to the parent plant unless mutations occur (Ağaoğlu et al., 2010). However, environmental factors like climate, soil conditions, and diseases may induce variations in the appearance or fruit quality of the plant. Clones, derived from a single parent plant through asexual propagation, exemplify this process (Soylu, 2003). Asexual propagation is employed for various reasons, notably to maintain desirable traits found in commercially valuable varieties. For instance, prized characteristics such as the flavor of a peach or the color of a strawberry can be preserved through cloning. In plants with parthenocarpic fruits, which develop without seeds, reproduction via seeds is not feasible. Therefore, asexual methods are essential for propagating fruit species and varieties incapable of producing viable seeds. Unlike plants grown from seeds, which possess unique genetic

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compositions, asexual reproduction ensures the retention of superior traits in horticultural plants vulnerable to genetic dilution through cross-pollination. Additionally, asexual propagation circumvents challenges posed by hard, impermeable seed coats, allowing for efficient plant regeneration (Ağaoğlu et al., 2010; Dumanoglu, 2021).

To obtain a new plant, portions of roots, stems, branches, or leaves from existing plants, known as "cuttings," are prepared. These cuttings are then encouraged to develop roots and shoots under optimal environmental conditions. Many fruit varieties such as quince, pomegranate, tea, olive, berry, vine, fig, and berry can be propagated directly through cuttings. Moreover, the use of cutting propagation is widespread in cultivating clonal rootstocks for fruit production and American grapevine rootstocks for viticulture, as well as for reproducing ornamental plants (Ağaoğlu, 2010). Given its ease and practicality, the cutting propagation method offers advantages over other forms of vegetative propagation, enabling the direct production of various seedling types and varieties (Ünal et al., 1992).

Stem cutting-based vegetative propagation offers a viable strategy to bolster the planting stock of *D. pilosanthera*, thereby aiding in its conservation efforts. Recognized as a swift method for generating genetically identical replicas of desired tree species, stem cutting propagation has proven effective for both commercial plantation and germplasm preservation endeavors (Shekhawat & Manokari, 2016; Bhuyan et al., 2017). Notably, research by the Department of Forestry (2019) underscores its significance in accelerating nursery management processes by expediting the production of robust root systems and ensuring high survival rates among cuttings from superior-quality trees. Given the imperative to swiftly produce superior planting stocks for indigenous species, particularly in light of competition from fast-growing exotics and genetically enhanced forest tree varieties, adopting efficient and cost-effective vegetative propagation techniques becomes paramount. Such methods hold promise for swiftly cloning and multiplying superior tree propagules, thereby bolstering reforestation and seed orchard initiatives with genetically uniform and high-quality planting materials.

(12) Scientific Basis/Theoretical Framework (Include concepts, theories and variables that indicate relevant relationships to be tested.

(13) Methodology (Details such as; a) variables to be measured and evaluated; b) treatments to be used and their layout; c) experimental procedures and design; d) statistical analysis; e) evaluation method and observations to be made, strategies for implementation.)

This study encompasses two distinct research investigations aimed at evaluating the survival rate and growth performance of various fruit trees and ornamental plants propagated through cuttings within a controlled clonal nursery environment. The first research segment focuses on assessing the viability and growth potential of different fruit tree species when propagated via cuttings, providing valuable insights into their adaptability to clonal propagation methods. Concurrently, the second research component delves into the survival rates and growth dynamics of diverse ornamental plant species propagated through cuttings, elucidating their responses to clonal propagation techniques. Through these complementary research endeavors, the study aims to contribute to the advancement of sustainable plant propagation practices and enhance our understanding of clonal nursery propagation methods for both fruit

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trees and ornamental plants.

Each study will be laid out in Complete Block Design for two factor experiment with three (3) Replications. Factor A comprised of length of cuttings, A₁ - 10 cm, A₂ -12.5 cm and A₃ - 15 cm. On the other hand, Factor B will be comprised of the various stages of stem: B₁ - Green, B₂ - Converging Green to Brown and B₃ - Brown. The experimental lay-out is shown below in figure 1.0:

A ₂ B ₂		A ₁ B ₁		A ₁ B ₃
A ₁ B ₁		A ₁ B ₂		A ₂ B ₃
A ₃ B ₁		A ₁ B ₃		A ₂ A ₂
A ₃ B ₂		A ₂ B ₃		A ₁ B ₁
A ₃ B ₃		A ₁ B ₃		A ₂ B ₁
A ₃ B ₂		A ₃ B ₃		A ₃ B ₁
A ₃ B ₃		A ₁ B ₂		A ₂ B ₁
A ₁ B ₂		A ₃ B ₁		A ₂ B ₁
A ₂ B ₃		A ₂ A ₂		A ₃ B ₂

Legend:

Factor A - Length of Cuttings

A₁ - 10 cm

A₂ - 12.5 cm

A₃ - 15 cm

Factor B - Stages of Stem

A₁ - Green

A₂ - Converging Green to Brown

A₃ - Brown

The bed will be prepared and sterilized meticulously to provide optimal growing conditions. Experimental plants will be spaced at 10 cm intervals across the bed, allowing enough space for growth and development.

Data to be gathered:

- 1. Plant Height (cm).** This parameter will be measured from the base of the plant to the top of the longest leaf using a tape measure. This will be collected at weekly interval.
- 2. Stem Girth (cm).** This parameter will be measured using a caliper.
- 3. Number of leaves.** This parameter will be taken by counting the total number of fully expanded leaves of the sample plants.
- 4. Number of Auxiliary Buds Sprout.** This parameter will be taken by counting the

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total number of Auxiliary Buds develop in each sample plants.

5. Length of roots (cm). This will be determined by measuring the length of roots of the sample plants in each treatments using a tape measure. This will be done at the termination of the experiment.

6. Survival Rate (%). This parameter will be determined by counting the number of plants that survived per replication and computed using the formula;

$$\text{Percentage of Survival} = \frac{\text{No. of plants that survived per replication}}{\text{Total no. of plants per replication}}$$

(14) Major Activities/Work plan (Enumerate in chronological order the task to be undertaken. Use gantt chart.

Research Objectives	Year 1				Year 2				Year 3				Year 4				Year 5			
	1 st Quarter	2 nd Quarter	3 rd Quarter	4 th Quarter	1 st Quarter	2 nd Quarter	3 rd Quarter	4 th Quarter	1 st Quarter	2 nd Quarter	3 rd Quarter	4 th Quarter	1 st Quarter	2 nd Quarter	3 rd Quarter	4 th Quarter	1 st Quarter	2 nd Quarter	3 rd Quarter	4 th Quarter
1. Proposal presentation																				
2. Revision and finalization of proposal																				
3. Acquisition of materials																				
4. Rehabilitation of Clonal Nursery																				
5. Conduct of the study and data gathering																				
6. Encoding of data for analysis																				
7. Writing of research manuscript																				
8. Presentation of Research result																				

(15) Expected Output (Indicate the specific products, processes or services which the project is expected to produce; quantify when possible)

The main objective of this study is to determine the most appropriate fruit and ornamental plants for propagation through cuttings. It aims to identify the ideal lengths and stages of stem growth for successful propagation. The study involves detailed experimentation and analysis to gather comprehensive insights into the optimal conditions for propagating a variety of trees and ornamental. The outcome of this research will be a well-researched brochure or magazine that serves as a reliable guide and fact-checking resource for those interested in propagating plants. This valuable publication will provide useful tips and accurate information, helping enthusiasts and horticulturists better understand the complex processes involved in successful propagation efforts.

(16) Technology Transfer Activity Design (Indicate the specific products, activity design, processes or services to be transferred as extension activities)

Product/Process/Service	Nature of Extension Activity	Target Beneficiaries
Training on Asexual Propagation	Technical, Advisory and Consultancy Services	Framers, Researchers, Students and Agricultural

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		Technologist
Quality Asexually Propagated Plants	Dispersal	Farmers

(17) Target Beneficiaries (Who the clientele are and what are the expected outcome/effects of the use of the projects output)

The findings of this research endeavor are poised to enrich a diverse spectrum of stakeholders, spanning from farmers and researchers to students and agricultural technologists. The wealth of knowledge and empirical data garnered through this study is anticipated to serve as a pivotal cornerstone for future explorations into asexual propagation, particularly with regard to cutting methodologies.

At its essence, this research aims to empower farmers with invaluable insights into the realm of asexual propagation. By delineating the trees and ornamental amenable to propagation via cuttings, this study endeavors to equip farmers with the requisite confidence, skills, and knowledge essential for the successful adoption of asexual propagation techniques. Moreover, this endeavor seeks to elucidate the optimal lengths and stages of stem development crucial for propagation success. Beyond its immediate objectives, this research is driven by a steadfast commitment to knowledge dissemination, seeking to impart practical expertise and insights on asexual propagation to farmers not only within the province but also across the wider region. Through this concerted effort, the research endeavors to catalyze positive transformations within the agricultural domain, championing sustainable practices and fostering tangible benefits for farmers and the broader community at large.

(18) Personnel Requirement

Name	Role (Program, Project, Study Leader)	Field of Specialization
Dr. Salvacion J. Legaspi	Study Leader	Plant Physiology
Kient Jerwin V. Tungala		

Percent time devoted to the Program/ Project/ Study

Program/Project Leader (in hours): 4 hours

Program/Project Members (in hours):

(19) Budgetary Requirement

SOURCES OF EXPENSES	Quantity	Amount (PhP)
A. Personnel Services (For studies needed the Research Assistant, informant/s, guide)	1	504.00/per day
B. Maintenance and Other Operating Expenses (MOOE) (modify and include only whichever is applicable)		250, 000.00
- Transportation allowance		20,000.00
- Meal allowance during data gathering		5,000.00
- Professional fees (laboratory analysis and other related fees)		
- Miscellaneous (for incidental expenses)		30, 000.00

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	- Printing		30, 000.00
	C. CAPITAL OUTLAY		
	Total		

(20) Literature Cited (include only those references published in reputable journals, and at least that from CAPSU Research Journal, <http://researchjournal.capsu.edu.ph/>)

Shekhawat, M. S. and M. Manokari. (2016). Impact of Auxins on Vegetative Propagation Through Stem Cuttings of *Couroupita guianensis* Aubl.: A Conservation Approach. [cited 2019 Oct 15]. Available from: <https://www.hindawi.com/journals/scientifica/2016/6587571/>

Bhuyan, L., C.L. Sharma, B. Singh and M.B. Sharma. (2017). Vegetative propagation of two important *Garcinia* species of Assam, NE India. *Journal of Medicinal Plants Studies*. 5(3): 273-277. Available from: <https://www.plantsjournal.com/archives/2017/vol5issue3/PartD/5-3-27-952.pdf>

Department of Forestry. (1997). Overview of Cutting Propagation. [cited 2019 Oct 15]. Available from: <http://www.rooting-hormone.com/hudson.html>

Oliveira, A. P.; Nienow, A. A.; Calvete, E. O. Capacidade de enraizamento de estacas semilenhosas e de cultivares de pessegueiro tratadas com aib. *Revista brasileira de fruticultura, jaboticabal*, v. 25, n. 2, p. 282-285, 2003.

Cardoso, C.; Yamamoto, I. Y.; Preti, E. A.; Assis, A. M.; Neves, C. S. V. J.; Roberto, S.R. Aibe substratos no enraizamento de estacas de pessegueiro ‘okinawa’ coletadas no outono. *Semina: Ciências agrárias, londrina*, v. 32, n. 4, p. 1307-1314, 2011

(21) Capsule Curriculum Vitae (One-page brief C.V. only)